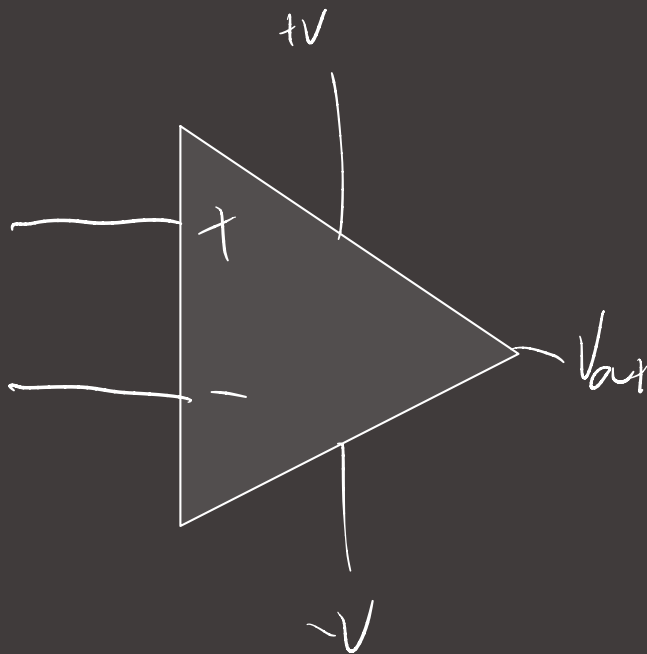


Chance Miller

OP-AMP Comparator

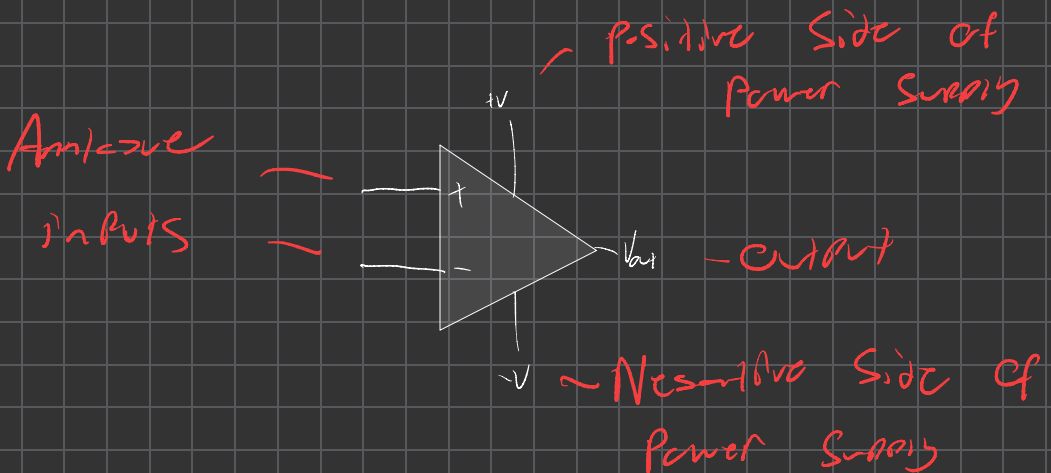


An OP-Amp (Operational Amplifier) is used for many things:

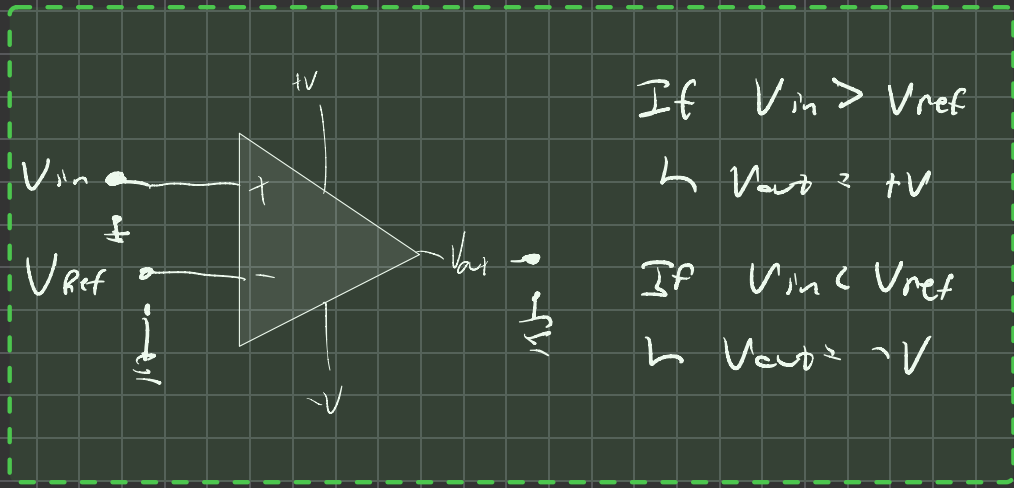
- DC Amplification

- Signal Filtering: 

- Mathematical operations ($+$, $-$, $\partial/\partial x$, $\int$)



One application of an OP-Amp is a Comparator, which takes two analog input voltages, and returns a 1 or 0 based on their comparison



The output is limited by the OP-Amp's Power Supply, that is, $V_{out} \nless V_{+}$, or $V_{out} \nless V_{-}$